



Building The Back End

A plain-English manual for the engine behind the online shop

Written so that anyone can follow it, whether or not they have ever built a website. Every technical word is explained the first time it appears.

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Subject: the YnR online shop (ynr-store.html)

How To Read This Document

This manual explains what still needs to be built for the YnR online shop, and just as importantly, why each piece matters. It is written for two readers at once:

- The person paying for the work, who needs to understand what they are buying and in what order it arrives.
- The person doing the work, who needs a checklist precise enough to build from.

Every section therefore follows the same shape. First a short explanation in everyday language, usually compared to a real shop. Then the actual list of things to build, numbered 1 to 57 straight through the document so any item can be pointed at directly in conversation (“we are on step 22”).

A NOTE ON THE NUMBERING

The numbers are a running order, not a rigid law. Steps inside a phase can be reshuffled. The phases themselves are deliberately ordered, because each depends on the one before it. You cannot record an order for a product the system does not yet know exists.

1. First, What Is A “Back End”?

Imagine a clothing shop on a real street.

The window display, the racks, the price tags, the mirror: that is the front end. It is everything a customer can see and touch. It is built to look good and to help someone choose.

Now think about everything the customer never sees. The storeroom where the stock actually sits. The notebook where the owner writes down who bought what. The drawer with the money in it. The list of phone numbers for regular customers. That is the back end.

THE ONE-SENTENCE VERSION

The front end is the shop window. The back end is the storeroom, the record book, and the till.

Here is the part that matters. When a customer points at a hoodie and asks “do you have this in medium?”, the assistant walks to the back to check, then returns with an answer. That walk to the back, and the answer carried out, is exactly what a back end does, thousands of times a day, in a fraction of a second.

A shop window with no storeroom behind it can show things. It cannot sell them.

2. Where The YnR Shop Is Today

The YnR website already exists, and it looks genuinely finished. That is worth saying plainly, because it makes the real problem easy to miss.

WHAT ALREADY WORKS

- It displays the six current pieces with photographs, prices in naira, and descriptions.
- A customer can open a piece, read about it, and choose a size.
- A customer can add pieces to a bag and see a running subtotal.
- Clicking checkout opens WhatsApp with the order already typed out, ready to send.

That is a real, working shop window. Someone could buy from YnR today using it.

WHAT IS ONLY PRETENDING TO WORK

Underneath, there is no storeroom at all. Six things are currently make-believe, and two of them are costing the business customers right now.

The newsletter sign-up is fake. When a customer types their email and clicks Sign Up, the page says “You’re on the list.” Nobody is on any list. There is no list. The address is discarded the instant they click, and nobody is told. This is a letterbox painted on a wall: post a letter into it, and the letter falls on the floor behind.

The contact form is fake. The same story. It says “Message sent.” No message goes anywhere. The customer’s name, email, subject and message are all deleted the moment the button is pressed. Someone could be asking to spend ₦75,000, and nobody at YnR will ever know they

wrote.

No order is ever written down. Checkout opens WhatsApp and then forgets everything. It is an assistant who walks the customer to the door, waves goodbye, and writes nothing in the book. A week later, nobody can answer “what exactly did this person order, and for how much?”

The bag empties when the page reloads. The shopping bag lives only in the browser’s short-term memory. Refresh the page, or come back tomorrow, and everything the customer chose is gone.

The six pieces are typed into the website’s code. They are written directly inside the page file. Adding a seventh piece, or changing a price, means editing code by hand. The shop cannot grow without a programmer on call.

Nothing knows what has been sold. Every YnR piece is one-of-one: there is exactly one of it in the world. Yet the system has no idea whether a piece is still available. This is the biggest gap, and section 5 explains why.

WHY THIS MATTERS MORE THAN IT SOUNDS

A form that quietly fails is worse than no form at all. If there were no newsletter box, nobody would expect anything. Because there is one, and because it says “You’re on the list,” real customers believe they will hear about the next drop. They will not. They will assume YnR forgot them.

3. The Words You Will Keep Hearing

Ten words cover almost everything in this document. None are complicated once the shop comparison is in place.

THE WORD	WHAT IT ACTUALLY MEANS
Server	A computer that is always switched on, sitting in a data centre, waiting to answer questions. It is the storeroom, plus the assistant who walks to it.
Database	The record book. A very well organised one, where every page has the same headings, so anything can be found quickly.
Table	One section of that record book. A Products table is the stock list. An Orders table is the sales ledger. Each has its own fixed columns.
Record	A single line in a table. One product. One order. One customer message.
Model	The blank form that decides what every record must contain. The Product model says every piece must have a name, a price, a category and a photograph. It is the printed form before anyone fills it in.
API	The serving hatch between the shop floor and the storeroom. The website is not allowed to rummage in the storeroom itself. It passes a request through the hatch and gets an answer back.
Endpoint	One specific window in that hatch, each with a single job. One window answers “list all products.” Another answers “tell me about this one piece.” A third accepts “here is a new order.”
Status	A label on a record saying where it has got to. A piece is available, held, or sold. An order is new, confirmed, paid, or delivered. Most of the useful work is moving these labels along correctly.
Admin panel	A private page, behind a password, where the owner of YnR manages everything without touching any code. The shopkeeper’s desk.
Webhook	A phone call coming the other way. Normally the website asks a question and waits for an answer. A webhook is an outside service, such as the bank, ringing YnR unprompted to say “that customer has just paid.”

4. The Build, Phase By Phase

Nine phases. Each explains what it is, why YnR specifically needs it, and exactly what gets built. The numbered steps run continuously from 1 to 57.

Phase 0. Foundations: the empty building

PICTURE IT

You cannot run a shop out of thin air. Before any stock arrives, you rent the building, put up shelves, connect the electricity, and cut a key for the front door. Nothing is for sale yet. But nothing can ever be for sale until this exists.

This phase produces nothing a customer will notice, which makes it tempting to skip. Skipping it is how a business ends up with its only copy of its sales records on one laptop that later gets stolen.

WHAT GETS BUILT

1. Choose the technology, set up the code repository, and separate the practice environment from the real one, so mistakes happen on a copy and never on the live shop.
2. Set up the database and a migration system, which is a controlled way to change the shape of the record book later without losing what is already written in it.
3. Arrange hosting, point the domain name at it, and switch on HTTPS, the padlock in the browser bar that stops outsiders reading what customers send.
4. Add logging and error handling, so that when something breaks there is a written trail explaining what happened instead of a shrug.
5. Set up automatic daily backups of both the database and the photographs, stored somewhere separate from the main server.

Phase 1. The stock list

PICTURE IT

Right now the six pieces are painted directly onto the shop window. To add a seventh, someone must repaint the window. The goal is to move them onto cards in a box, so adding one is just writing a new card.

This is where the six hardcoded pieces come out of the website's code and into a real, searchable list. Once it is done, the website stops knowing anything about specific products, and instead asks the storeroom "what have we got?" each time someone visits.

WHAT GETS BUILT

6. The Product model: web address name, display name, category, price, description, available sizes, status, and the date it was added.
7. The Category model, covering the four YnR categories: Tees, Hoodies, Long Sleeves and Denim.

8. The ProductImage model, so one piece can have several photographs in a chosen order, each with a text description for screen readers and search engines.
9. Photograph upload and storage, with automatic resizing, so a large camera image does not make the page slow to load on a phone.
10. An endpoint that lists all products, with filtering by category and control over display order.
11. An endpoint that returns one single piece in full detail.
12. An endpoint that returns related pieces, for the “You May Also Like” row.
13. A seeding script that loads the existing six pieces in automatically, so nobody has to retype them.

Phase 2. The “held” sticker: one-of-one stock

PICTURE IT

Two customers message on WhatsApp about the same hand-painted hoodie within the same ten minutes. There is only one hoodie. If both are told yes, one of them is about to be disappointed, and YnR looks unreliable through no fault of its own.

This phase is the heart of the system, because it is the one thing about YnR that an ordinary online shop never has to solve. A shop selling factory t-shirts holds two hundred of each. If two people buy at once, no problem. YnR has exactly one of everything. There is no second copy, ever.

The answer is a temporary hold. When a customer begins ordering, the piece gets a “held” sticker for a set time, perhaps thirty minutes. During that window nobody else can claim it. If the customer completes the order, it becomes sold. If they wander off, the sticker falls off by itself and the piece returns to the shop.

The site’s own words also promise something specific: sold pieces stay visible, and customers can ask for one to be repainted. That promise needs somewhere to live.

WHAT GETS BUILT

14. A status system for every piece, moving through available, held, sold, and repaint-only, with clear rules about which moves are allowed.
15. The hold itself, reserving a piece for a fixed number of minutes while a customer is ordering.
16. A background job that runs continuously and clears expired holds, returning abandoned pieces to the shop without anyone doing it by hand.
17. The sold archive, so a sold piece stays on the site as proof of the work rather than disappearing.
18. A repaint request feature, letting a customer ask for a fresh version of a piece that has already gone.
19. Availability tracked per size, for any piece where different sizes genuinely exist as separate physical items.

Phase 3. The record book: orders

PICTURE IT

A customer fills a bag, clicks checkout, and WhatsApp opens with their order neatly typed out. Then the website forgets they ever existed. Nothing is written down. If that chat is ever deleted, the order goes with it.

This is almost certainly the highest-value work in the entire document, because YnR currently has no sales records of any kind beyond a phone's chat history.

The key change is small but crucial. The order is written into the record book first, and only then does WhatsApp open. The message that opens carries an order reference, something like YNR-4417, so the chat and the record are tied together permanently.

THE SNAPSHOT RULE

Prices must be frozen into the order at the moment it is placed. If the Icon Hoodie sells at ₦75,000 today and rises to ₦85,000 next month, an order from today must still say ₦75,000 for ever. Storing a link to the current price, instead of a copy of it, silently rewrites history. It is one of the most common and most damaging mistakes in shop software.

WHAT GETS BUILT

- 20.** The Order model: reference code, customer details, status, subtotal, delivery fee, total, and which channel the order arrived through.
- 21.** The OrderItem model, holding a frozen copy of the name, size and price of each piece at the time of ordering.
- 22.** An endpoint that creates the order record before the customer is handed over to WhatsApp.
- 23.** WhatsApp links generated by the server, carrying the order reference inside the message.
- 24.** The order journey: new, awaiting confirmation, confirmed, paid, being painted, shipped, delivered, plus cancelled at any point.
- 25.** A lookup endpoint, so a customer can check their own order using its reference, with no account and no password.
- 26.** A saved bag, so a customer's choices survive closing the page and returning later.
- 27.** An instant notification to the owner whenever a new order is placed.

Phase 4. The map and the price list: delivery

PICTURE IT

The delivery prices are currently painted on the shop wall. Changing the Abuja rate means calling a programmer to repaint it.

YnR delivers on three tiers: a flat ₦2,000 inside Abuja, a variable rate for the rest of Nigeria, and courier rates internationally. Those numbers sit in the website's code today, which means the owner cannot adjust them.

WHAT GETS BUILT

- 28. A delivery zone model covering Abuja, the rest of Nigeria, and international, each with its own pricing rule.
- 29. A fee calculation endpoint, so the correct delivery cost is worked out by the system rather than typed by hand.
- 30. Address capture and checking, so parcels are not lost to a mistyped street.
- 31. Courier tracking references stored against the order, with status updates the customer can see.

Phase 5. The letterbox: messages that actually arrive

PICTURE IT

A letterbox painted on a wall. Customers post letters into it every week. The letters fall on the floor behind and are swept up at closing time. The customers never find out.

This phase stops the website misleading people. Both forms currently show a success message and then throw everything away. Of all the work in this document, this is the smallest and the most urgent, because it is losing real customers today, silently, with no way of knowing how many.

WHAT GETS BUILT

- 32. A contact endpoint that genuinely saves the message, alerts the owner, and sends the customer a real acknowledgement.
- 33. A ContactMessage model, and a simple inbox to read and reply from.
- 34. A newsletter sign-up endpoint with email confirmation, so an address is only added once its owner has clicked to confirm it.
- 35. A Subscriber model with working unsubscribe links, which is both good manners and a legal requirement.
- 36. A connection to an email delivery service, so messages reliably reach inboxes instead of spam folders.
- 37. A drop announcement tool, letting the owner tell the whole list at once when a new one-of-one piece goes up.
- 38. Optional: a proper WhatsApp Business connection, which would let order confirmations be sent automatically rather than typed out by hand.

Phase 6. The shopkeeper's desk: the admin panel

PICTURE IT

Everything so far has been built for the computer. This phase is the only part built for the human being who actually runs YnR.

This phase decides whether the shop can survive without a programmer. If adding a newly painted piece needs someone technical, the shop stops whenever that person is busy, travelling, or no longer working with YnR. The admin panel is what makes the business independent.

It should be simple enough to use on a phone, standing in the shop, one-handed, while holding a freshly painted hoodie in the other.

WHAT GETS BUILT

- 39. A secure login, so only the owner can reach the desk.
- 40. Full product management: add, edit, remove and reorder pieces, with drag-and-drop photograph upload.
- 41. A single-tap control to mark a piece sold, or available again.
- 42. An order board showing every order and allowing it to be moved along its journey.
- 43. The message inbox and the subscriber list, together in one place.
- 44. A simple dashboard: orders waiting, money taken, and which sold pieces are most often asked to be repainted.
- 45. An activity log recording who changed what, and when.

Phase 7. The till: taking payment online

PICTURE IT

Right now all money is handled by conversation on WhatsApp, which in Nigeria is completely normal and works perfectly well. This phase is genuinely optional, and it sits late in the order on purpose.

Do not build this before the phases above it. An online card system on top of a shop that cannot yet record an order is expensive decoration. Once orders exist, payment is straightforward to add.

WHAT GETS BUILT

- 46. Integration with Paystack or Flutterwave, both built for Nigeria and both handling naira natively.
- 47. Payment links tied to a specific order reference, so money and order can never be mismatched.
- 48. A webhook handler with signature checking, so the system believes a “they have paid” message only when it genuinely came from the bank.
- 49. Bank transfer support with proof-of-payment upload, which remains how a great many Nigerian customers prefer to pay.
- 50. Reconciliation, refunds and receipts.

Phase 8. Locks on the doors: security and safety

PICTURE IT

A shop with beautiful stock, a full till, and no lock on the front door.

Public forms attract automated abuse within days of going live, not months. This phase is not optional, and much of it is quick.

WHAT GETS BUILT

- 51.** Checking of everything a customer sends, done on the server, never trusting the browser to have checked it already.
- 52.** Rate limiting, so no single person or robot can hammer the forms thousands of times.
- 53.** Spam protection, starting with a hidden trap field that only robots fill in, adding a puzzle only if that proves insufficient.
- 54.** Standard web security protections and headers.
- 55.** Compliance with the Nigeria Data Protection Act 2023: a privacy policy, clear consent, and a way for customers to see or delete the data held about them.
- 56.** Monitoring that alerts someone when the site goes down, rather than waiting for a customer to mention it.
- 57.** Analytics showing which pieces get viewed, how many viewers click through to WhatsApp, and how many of those become orders.

5. The One Genuinely Hard Problem

Almost everything in this document is ordinary work. An experienced developer has built each of these pieces before. One thing, though, deserves a decision made deliberately and early, because getting it wrong is visible to customers and awkward to fix afterwards.

It is the hold, from Phase 2.

Because YnR sells one-of-one pieces confirmed by a human being over WhatsApp, there is an unavoidable gap between a customer saying “I want this” and the owner replying “yes, it is yours.” That gap might be two minutes. Overnight, it might be nine hours. For that entire window the piece is in limbo: not sold, but not really available either.

Three questions settle the design, and all three are business decisions rather than technical ones:

- How long does a hold last before the piece returns to the shop? Too short, and genuine customers lose pieces they were still deciding on. Too long, and the shop looks sold out when it is not.
- What does another customer see while a piece is held? “Sold”? “Reserved”? Or nothing at all, letting them enquire anyway and be told if it frees up?
- Can the owner override a hold from the admin panel? Almost certainly yes, but it needs building deliberately, not discovered in a panic.

DECIDE THIS BEFORE ANY ORDER CODE IS WRITTEN

Holds touch the product system, the order system and the admin panel all at once. Changing the answer later means changing all three. Ten minutes of conversation now saves a week of rework.

6. If Only Three Things Get Built

Fifty-seven steps is a real project, likely several months of part-time work. If the choice is between doing all of it eventually and doing something useful within a fortnight, these three deliver most of the value, and each can be built independently of the rest.

First: make the two forms real (steps 32 and 34). This is the smallest job in the document, and the only one actively costing money today. Every day it waits, more customers believe they have contacted YnR when they have not.

Second: record orders before the WhatsApp handoff (steps 20 to 23). This turns a chat history into an actual business record. It also means that if a phone is ever lost or stolen, the orders are not lost with it.

Third: move the products into a database with a sold flag (steps 6 to 13, plus 14). This is what lets YnR add a newly painted piece without a programmer, and stops the shop offering things that are already gone.

Payment, international delivery rules, analytics and the full admin dashboard can all wait. None of them matter if an order cannot be written down.

7. Questions To Answer Before Anyone Writes Code

Every one of these is a business decision, not a technical one. A developer cannot guess them, and guessing wrong means rework. Answering them takes an afternoon.

- How long should a piece stay held while a customer is deciding?
 - When a piece sells, does it stay visible on the site? The current site copy promises that it does, so this needs confirming.
 - Who has admin access, and what happens to that access if they leave?
 - Should online card payment exist at launch, or is WhatsApp-confirmed payment enough for now?
 - Whose WhatsApp number receives orders, and does the business own that number rather than an individual?
 - How is the international courier rate worked out, and who quotes it?
 - For repaint requests, is there a standard waiting time and price, or is each one negotiated?
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8. One Thing To Fix On The Website Itself

This is not back end work, but it was found while reading the site's code and should not be lost.

The site contains a complete, well-built size guide, with chest and length measurements covering S through XXL. It is finished and it looks good. But no button anywhere on the website opens it. The code that would display it is never triggered by anything, so no customer can ever see it.

Given that sizing is one of the most common reasons a customer hesitates or stops to ask a question, connecting a "Size Guide" link on the product page is perhaps ten minutes of work for a noticeable gain.

End of manual. Steps are numbered 1 to 57 for easy reference in discussion.